

A tour of literate scripts

This page exercises what a report needs: interpolated numbers, a table built by a comprehension, math, footnotes, admonitions, a cached figure, and an early stop that leaves the rest of the document readable. Cell code is hidden, as it is by default: a report hides its plumbing. A page that is about the code, like the themed-plots one, opts in with `# code: show` at the top of the file.

The file is plain Python, so `ruff`, `ty`, and the IDE see every cell. A top-level string is prose (this paragraph is one), and the code between two of them is a cell.

Numbers in prose

The best condition is `anchored` at 0.531 ± 0.021 over 20 seeds, against a gate of 0.45. Prose that quotes a value is an f-string, so a format spec does what it does anywhere in Python, and the names are real references that ruff, ty, and go-to-definition see; a helper in a cell can do anything more involved.

A table from a comprehension

Today a table like this is assembled as a Markdown string in Python.

Here the rows are one field, written where the table appears:

condition	seeds	mean	verdict
control	20	0.012 ± 0.004	miss
anchored	20	0.531 ± 0.021	pass
anchored-slot	12	0.498 ± 0.030	pass

Math, footnotes, admonitions

The same Markdown extensions as `mo.md` : inline math $\bar{\alpha} =$

$\frac{1}{n} \sum_i \alpha_i$, display math

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \|h - a\|^2,$$

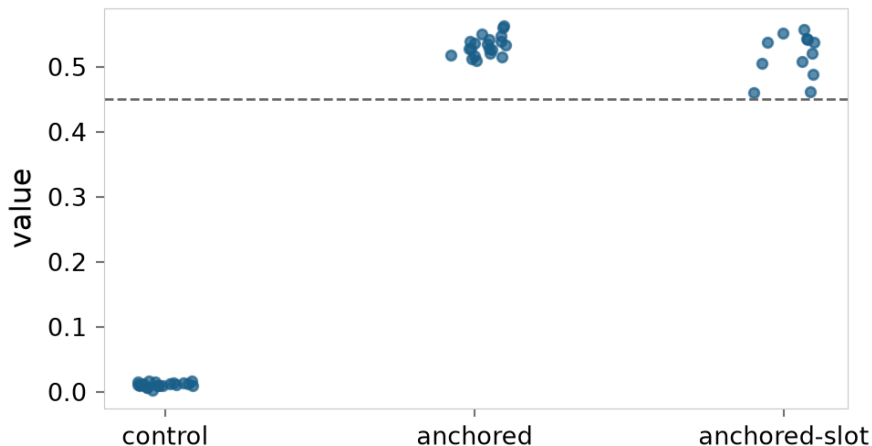
a footnote¹, and an admonition. This paragraph is a plain raw string, since the math is full of braces; the admonition below quotes a value, so it is a separate f-string (a paragraph that needs both doubles its math braces).

A note on the gate

The gate of 0.45 comes from the reference experiment, quoted here from the same constant the cells use.

A cached figure

The slow part of a re-render is usually the figures. `@memo` caches the rendered HTML keyed by the plot function's source and its arguments (arrays included), and remembers the PNGs it wrote, so the next render, even in a fresh process, skips the draw. This one sleeps for a second to make the point.



Per-seed values by condition. The dashed line is the gate; each dot is one seed.

Stopping early

A preregistration is written before its results exist. `stop()` ends execution with a message, and every later field that cannot be evaluated renders as a pending mark instead of an error, so the whole document still reads: headings, paragraphs, and the values that are known.

Results to come

The experiment has not been published yet.

Results. The anchored condition reached `summary.m_line` with a lead of `summary.lead`; see `summary.figure`.

This paragraph has no interpolation, so it renders as written.

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1. Footnotes number from one per document, since the whole page is rendered in one pass. ↩